Experiment No:11

Experiment to understand pandas library use cases in Data science

AIM:

To understand pandas library use cases in Data science.

DESCRIPTION:

Understand data frame use cases using pandas library

CODE:

```
import numpy as np
import pandas as pd


# Create a DataFrame using a list
list = [[1, 'Smith', 50000],
        [2, 'Jones', 60000]]


df = pd.DataFrame(list)
df
# Rename the columns
df.columns = ['Empd', 'Name', 'Salary']
df
# Display DataFrame information
```

```
df.info()


# Read Student Marks dataset

df = pd.read_csv('/content/Student_Marks.csv')

df.info()

# Display first five rows

df.head()

# Display last five rows

df.tail()

# Read Housing dataset

df = pd.read_csv('/content/housing.csv (1).zip',
compression='zip')

# Display first five rows

df.head()

# Display last five rows

df.tail()

# Display dataset information

df.info()

# Display median house value

df['median_house_value']

# Display data type
```

```
type(df['median_house_value'])

# Calculate mean

df['median_house_value'].mean()

# Calculate median

df['median_house_value'].median()

# Calculate mode

df['median_house_value'].mode()

# Calculate variance

df['median_house_value'].var()

# Calculate standard deviation

df['median_house_value'].std()

# Statistical summary

df.describe()

# Statistical summary including categorical columns

df.describe(include='all')

# Get column names

empCol = df.columns

empCol

# Convert DataFrame into NumPy array

emparray = df.values

emparray
```

```
# Convert NumPy array back into DataFrame
employee_DF = pd.DataFrame(emparray, columns=empCol)
employee_DF
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
```

Data columns (total 3 columns):

#	Column	Non-Null Count	Dtype
0	number_courses	100 non-null	int64
1	time_study	100 non-null	float64
2	Marks	100 non-null	float64

dtypes: float64(2), int64(1)

memory usage: 2.5 KB

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 20640 entries, 0 to 20639

Data columns (total 10 columns):

#	Column	Non-Null Count	Dtype
0	longitude	20640 non-null	float64
1	latitude	20640 non-null	float64
2	housing_median_age	20640 non-null	float64
3	total_rooms	20640 non-null	float64
4	total_bedrooms	20433 non-null	float64
5	population	20640 non-null	float64
6	households	20640 non-null	float64
7	median_income	20640 non-null	float64
8	median_house_value	20640 non-null	float64
9	ocean_proximity	20640 non-null	object

dtypes: float64(9), object(1)

memory usage: 1.6+ MB

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity
0	-122.23	37.88	41.0	880.0	129.0	322.0	126.0	8.3252	452600.0	NEAR BAY
1	-122.22	37.86	21.0	7099.0	1106.0	2401.0	1138.0	8.3014	358500.0	NEAR BAY
2	-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100.0	NEAR BAY
3	-122.25	37.85	52.0	1274.0	235.0	558.0	219.0	5.6431	341300.0	NEAR BAY
4	-122.25	37.85	52.0	1627.0	280.0	565.0	259.0	3.8462	342200.0	NEAR BAY
...	...	...	...	...	...	...	...	...	...	...
20635	-121.09	39.48	25.0	1665.0	374.0	845.0	330.0	1.5603	78100.0	INLAND

20636	-121.21	39.49	18.0	697.0	150.0	356.0	114.0	2.5568	77100.0	INLAND
20637	-121.22	39.43	17.0	2254.0	485.0	1007.0	433.0	1.7	92300.0	INLAND
20638	-121.32	39.43	18.0	1860.0	409.0	741.0	349.0	1.8672	84700.0	INLAND
20639	-121.24	39.37	16.0	2785.0	616.0	1387.0	530.0	2.3886	89400.0	INLAND

20640 rows × 10 columns

RESULT:

Thus, the Pandas library was studied and implemented successfully to understand its use cases in Data Science, including creating DataFrames, reading CSV and ZIP datasets, displaying dataset information, performing statistical operations, and converting DataFrames into NumPy arrays and back